Math 152 Lecture Notes: Week 5

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0.1 Volume as the integral of cross sectional area

Last week we introduced the integral formula for the volume of a solid:

$$V = \int_a^b A(x)\mathrm{d}x$$

Here the solid extends from $x = a$ to $x = b$, and $A(x)$ represents the cross-sectional area of the solid at x, the area of the plane region obtained by slicing the solid with a plane perpendicular to the x-axis. Any axis can play the role of the x-axis in computing volume; as with area, you should choose an axis with an eye towards finding a formula for the cross sectional area.

Example 1. Find the volume of a structure extending from $x = 1$ to $x = 4$ whose cross sections perpendicular to the x-axis are the regions below parabolic arcs $y = x - z^2$. [Draw picture]

First we need to find the cross sectional area $A(x)$; it makes sense to use x as we are given a description of these cross sections. Fixing x, the cross section extends from $z = -\sqrt{x}$ to $z = \sqrt{x}$ [picture], and so our cross sectional area is:

$$A(x) = \int_{-\sqrt{x}}^{\sqrt{x}} [x - z^2]\mathrm{d}z = 2\int_0^{\sqrt{z}} [x - z^2]\mathrm{d}z = 2\left[xz - \frac{z^3}{3}\right]\Big|_0^{\sqrt{x}} = 2x^{3/2} - 2\frac{x^{3/2}}{3} = \frac{4x^{3/2}}{3}$$

Now we can find the volume

$$V = \int_1^4 A(x)\mathrm{d}x = \int_1^4 \frac{4x^{3/2}}{3}\mathrm{d}x = \left[\frac{4}{3}\cdot\frac{2}{5}x^{5/2}\right]\Big|_1^4 = \frac{8}{15}\left[32 - 1\right] = \frac{31\cdot 8}{15}$$

1 Solids of Revolution, Part II

1.1 Disks and Washers

At the end of last week, we saw that for solids with rotational symmetry:

1. We can describe them as *solids of revolution*, swept out by a plane region as it rotates about an axis.
2. Their cross sections are typically[1] disks or "washers" (annuli), and so we can obtain integral formulas for their volume given a description of curves bounding the region being revolved.

[1] I saw typically because more generally one could have something like a collection of concentric washers [picture].

In particular, if the region bounded below by $f(x)$ and above by $g(x)$ over $[a,b]$ is revolved around the x-axis, the volume of the resulting solid is

$$\int_a^b \pi[g(x)^2 - f(x)^2]\mathrm{d}x$$

Here $g(x)$ represents the outer radius of the cross section, $f(x)$ the inner radius. As before any axis can play the role of the x-axis. If the region being revolved abuts the axis of rotation [picture] (so the solid has no hole through it) then $f = 0$, and we only need to consider the outer radius.

Example 2. Consider the solid obtained by revolving the region between the graphs of $x = 1 + y$ and $x = 3$ and between $y = 0$ and $y = 2$ about the y-axis. [Draw picture]. The cross sections are washers with inner radius $1 + y$ and outer radius 3, so the volume is

$$\int_0^2 \pi(3^2 - (1+y)^2)\mathrm{d}y = 18\pi - \pi\int_0^2 (1+y)^2\mathrm{d}y = 18\pi - \pi\int_1^3 u^2\mathrm{d}u$$

$$= 18\pi - \pi\left[\frac{u^3}{3}\right]\Bigg|_1^3 = 18\pi - \pi(9 - 1/3)$$

Example 3. Let's use this approach to find the volume of a cone with base radius r and height h. This is obtained by revolving a triangle with vertices $(0,0), (r,0)$ and $(0,h)$ around the y-axis. We are integrating in y, so we need to express x as a function of y along the hypotenuse of this triangle; this will give us the outer radius. This is the line passing through $(0,h)$ and $(r,0)$, so it has equation

$$y = -\frac{h}{r}x + h$$

or equivalently

$$x = -\frac{r}{h}y + r = r(1 - y/h)$$

Thus our integral for volume is

$$\int_0^h \pi r^2(1-y/h)^2\mathrm{d}y = -\int_1^0 \pi r^2 h(1-u)^2\mathrm{d}u = \pi r^2 h\int_0^1 u^2\mathrm{d}u = \frac{\pi r^2 h}{3}$$

1.2 Shells

By considering cross sections perpendicular to the axis of rotation, we have seen how to find the volume of a solid obtained by revolving the regions between the graphs of functions $f(x)$ and $g(x)$ around the x-axis. It is not so clear how we could find the volume of the solid obtained by revolving this same region about the y-axis, without explicitly finding right and left boundary curves for the region. [Draw picture with region between parabolas].

There is an alternative way to find the volume of solids of revolution like this, which comes out of a different way of approximating volumes. Where before we approximated volume using a stack of extrusions of flat cross sections, we will now approximate it using concentric cylindrical shells.

A cylindrical shell is just an extrusion of an annulus (washer) [draw picture.] The volume of such a solid is $\pi(R^2_{\text{outer}} - R^2_{\text{inner}})h$. We will be considering very thin cylindrical shells, so it is convenient to describe them setting the inner average of the inner and outer radii as r and width of the shell as Δr. Then the inner and outer radii are $r + \frac{\Delta r}{2}$ and $r - \frac{\Delta r}{2}$, and so the volume comes out to $2\pi \cdot r \cdot \Delta r \cdot h$.

Returning to the general problem of finding the volume of a solid of revolution, suppose we revolve the region between $f(x)$ and $g(x)$ over the interval $[a,b]$ around the y-axis. [Draw picture with relatively simple

curves.] If we take a partition of $[a, b]$ into subintervals $[x_i, x_{i+1}]$ of width Δx_i, we can form a collection of cylindrical shells with inner and outer radii given by x_i and x_{i+1}. Letting $t_i = \frac{x_i + x_{i+1}}{2}$ be the midpoint of the ith interval we approximate the corresponding region of the solid by setting the height of the cylindrical shell equal to $h(t_i) = g(t_i) - f(t_i)$. [Draw picture] Summing these up, the total volume of the solid is approximated by

$$\sum_i 2\pi \cdot t_i \cdot f(t_i) \cdot \Delta x_i$$

This is a Riemann sum for the function $2\pi x h(x)$ on the interval $[a, b]$. Letting our partition get finer and finer to better approximate the volume, this Riemann sum approaches

$$\int_a^b 2\pi x h(x) \mathrm{d}x$$

as well as the true volume.

Conclusion. *If we revolve a plane region between $x = a$ and $x = b$ with cross sectional height $h(x)$ about the y-axis, the resulting solid has volume*

$$\int_a^b 2\pi x h(x) \mathrm{d}x$$

Remark. Here is another point of view on this formula. The quantity inside the integral, $2\pi x h(x)$, represents the area of the surface of a cylinder with height $h(x)$ and radius x. Thus instead of integrating the areas of flat cross sections, we are integrating the areas of concentric cylindrical sections of the solid. [Picture]

Example 4. Let us again compute the volume of a cone with base radius r and height h, to make sure we get the same answer as with the disk method. As before, we are revolving the triangle below

$$y = -\frac{h}{r}x + h$$

about the y-axis. The shell method gives us the following integral formula for the volume:

$$V = \int_0^r 2\pi x \left[-\frac{h}{r}x + h\right] \mathrm{d}x = \int_0^r \left[2\pi h x - \frac{2\pi h}{r}x^2\right] \mathrm{d}x = \left[\pi h x^2 - \frac{2\pi h}{3r}x^3\right]\Big|_0^r = \pi h r^2 - \frac{2\pi h r^2}{3} = \frac{\pi r^2 h}{3}$$

Example 5. Consider the solid obtained by revolving the region between $y = x^2 + 1$ and $y = 2x^4$ about the y-axis. [Draw picture, with intersection points.] The relevant plane region is between $x = 0$ and $x = 1$, and our integral using the shell method is

$$\int_0^1 2\pi x[x^2 + 1 - 2x^4] \mathrm{d}x = \int_0^1 \left[2\pi x^3 + 2\pi x - 4\pi x^5\right] \mathrm{d}x = \left[\frac{\pi x^4}{4} + \pi x^2 - \frac{4\pi x^6}{6}\right]\Big|_0^1 = \frac{\pi}{4} + \pi - \frac{2\pi}{3} = \frac{7\pi}{12}$$

As with our other methods, we can use the shell method for solids of revolution about any axis, provided that we have a formula for the cross sectional height/width of the plane region at a given distance from the axis. [Draw picture with a cylindrical section of a solid of revolution about the x-axis.]

Example 6. Consider the quadrilateral in the plane with vertices $(1, 1), (2, 1), (2, 2), (4, 2)$. [Draw picture]. The left and right sides are the lines $x = y$ and $x = 2y$. We will compute the volume of the solids obtained by revolving this about the x and y axes, respectively. Because we have nice expressions for the left and right endpoints of the region as a function of y, it makes sense to evaluate the volume of the solid of revolution about the x-axis using the shell method, and the solid of revolution about the y-axis using the washer method.

Let's start with the solid of revolution about the x-axis. [Draw picture of cylindrical shell]. The width of the solid at height y is $2y - y = y$, and so the shell method gives us

$$\int_1^2 2\pi y^2 \mathrm{d}y = \left[\frac{2\pi y^3}{3}\right]_1^2 = \frac{14\pi}{3}$$

Now let's consider the solid of revolution about the y-axis. Our inner radius at y is y, and our outer radius is $2y$, so the washer method gives us

$$\int_1^2 \pi((2y)^2 - y^2)\mathrm{d}y = \int_1^2 3\pi y^2 \mathrm{d}y = \left[\pi y^3\right]\Big|_1^2 = 7\pi$$

This concludes our discussion of volume.

2 Averages

We will now consider another interpretation of the integral, in terms of average values.

2.1 The Average Value of a Function

Our interpretation of the integral in terms of averages will go through the fundamental theorem of calculus and the notion of the average rate of change of a function. Recall that for a function $F(x)$, the average rate of change of F between a and b is

$$\frac{F(b) - F(a)}{b - a}$$

This is the slope of the secant line between $(a, F(a))$ and $(b, F(b))$.

Example 7. The average velocity of an object between $t = a$ and $t = b$ is its displacement $x(b) - x(a)$ divided by the length of time $b - a$.

How is this quantity related to the instantaneous rate of change F'? Well given $F'(x)$, the fundamental theorem of calculus tells us how to recover the change in f over an interval.

Proposition (Accumulated change formula)**.**

$$F(b) - F(a) = \int_a^b F'(t)\mathrm{d}t$$

Indeed, F is an antiderivative of F' by definition.

Example 8. Suppose that during a storm between $t = 0$ and $t = 2$ hours, rain falls at a rate of $1 - (t-1)^2$ inches per hour at time t. [Draw graph]. What is the total precipitation during the storm?

This is the integral

$$\int_0^2 [1 - (t-1)^2]\mathrm{d}t = \int_{-1}^1 [1 - u^2]\mathrm{d}u = 2\int_0^1 [1 - u^2]\mathrm{d}u = 2\left[u - u^3/3\right]\Big|_0^1 = 4/3$$

Using this, the average rate of change can be written

$$\frac{1}{b-a}\int_a^b F'(x)\mathrm{d}x$$

Now given any continuous function f, f is the rate of change of its accumulation function

$$F(x) = \int_a^x f(t)\mathrm{d}t$$

so it makes sense to say that the average value of f on $[a, b]$ is

$$\frac{1}{b-a}\int_a^b f(x)\mathrm{d}x$$

This will be our definition of the average value of a function.

Definition. Given a function f on an interval $[a, b]$, its average value on this interval is

$$\overline{f} = \frac{1}{b-a}\int_a^b f(x)\mathrm{d}x$$

Example 9. What is the average value of $f(x) = x$ on $[a, b]$? This is

$$\frac{1}{b-a}\int_a^b x\mathrm{d}x = \frac{1}{b-a}\left(\frac{b^2-a^2}{2}\right) = \frac{a+b}{2}$$

as expected.

Example 10. What is the average value of $f(x) = x^n$ on $[0, 1]$? This is

$$\frac{1}{1-0}\int_0^1 x^n\mathrm{d}x = \frac{1}{n+1}$$

For all n, the function x^n ranges from 0 to 1 on this interval. As n grows larger, the function "spends more time" having a small value, so its average value decreases. [Draw pictures of curves].

Example 11. What is the average value of the sine function on $[0, 2\pi]$? on $[0, \pi]$? The first is

$$\frac{1}{2\pi}\int_0^{2\pi} \sin(x)\mathrm{d}x = \frac{1}{2\pi}(\cos(0) - \cos(2\pi)) = 0$$

The second is

$$\frac{1}{\pi}\int_0^{\pi} \sin(x)\mathrm{d}x = \frac{\cos(0) - \cos(\pi)}{\pi} = \frac{2}{\pi}$$

Now is a good time to discuss integration of some mildly discontinuous functions, so that we can discuss their average values.

Definition (Piecewise continuous)**.** I will say that a function is *piecewise continuous* on an interval $[a, b]$ if there is a partition of the interval into sub-intervals $[x_i, x_{i+1}]$ such that f is continuous on each open interval (x_i, x_{i+1}) and has well defined left and right limits at the endpoints [draw picture].

Definition. Given $f : [a, b]$ piecewise continuous, we define

$$\int_a^b f(x)\mathrm{d}x = \sum_i \int_{x_i}^{x_{i+1}} f(x)\mathrm{d}x$$

where the x_i are chosen so that f is continuous on each (x_i, x_{i+1}).

Example 12. The piecewise function

$$f(x) = \begin{cases} -1 & x \in [0,1] \\ x & x \in (1,3] \end{cases}$$

has average value

$$\frac{1}{3}\int_0^3 f(x)\mathrm{d}x = \frac{1}{3}\left[\int_0^1 [-1]\mathrm{d}x + \int_1^3 x\mathrm{d}x\right] = \frac{1}{3}\left[-1+4\right] = 1$$

Notice that this is the weighted average of -1 and 2, the average values of f on $[0,1]$ and $[1,3]$, weighted by the lengths of these intervals.

In general, given $a < b < c$, the average of a function f on $[a,c]$:

$$\overline{f}_{[a,c]} = \frac{1}{c-a}\int_a^c f(x)\mathrm{d}x$$

can be written as a weighted average

$$\frac{b-a}{c-a}\overline{f}_{[a,b]} + \frac{c-b}{c-a}\overline{f}_{[b,c]}$$

Indeed, one can define the average as a limit of weighted averages of sample values in a partition; these become Riemann sums for the integral defining the average.

Read backwards, our definition of average gives us the following interpretation of the integral.

$$\int_a^b f(x)\mathrm{d}x = \overline{f}\cdot(b-a)$$

[Draw picture with area under curve]

We can interpret our integral formulas for volume and area in light of this:

$$\text{Area} = \text{width}\cdot\text{average cross sectional height} = \text{height}\cdot\text{average cross sectional width}$$

$$\text{Volume} = \text{width}\cdot\text{average cross sectional area} = (\text{outer radius}-\text{inner radius})\cdot\text{average area of cylindrical cross section}$$

2.2 The Mean Value Theorem for Integrals

A useful fact about continuous functions is that they are guaranteed to attain their average value at some point.

Theorem 1 (Mean Value Theorem for Integrals)**.** *Let f be continuous on $[a,b]$. Then there is a point $x_0 \in [a,b]$ such that*

$$f(x_0) = \overline{f} = \frac{1}{b-a}\int_a^b f(x)\mathrm{d}x$$

Example 13. If a solid has width 5 and volume 10, the some cross section of it must have area exactly 2, provided that its cross sectional area varies continuously, without sharp "steps." [Draw picture]

Continuity is necessary for this to be true, the function

$$f(x) = \begin{cases} 0 & x \in [0,1] \\ 2 & x \in (1,2] \end{cases}$$

has average value 1 on $[0,2]$, but never attains the value 1.

To prove this, we will use another useful property of continuous functions.

Theorem 2 (Intermediate Value Theorem)**.** *Suppose that f is continuous on an interval $[a, b]$, and that $f(a) \leq c \leq f(b)$ or $f(b) \leq c \leq f(a)$. Then there is an $x \in [a, b]$ such that $f(x) = c$. [Draw pictures]*

Proof of the mean value theorem. Because f is continuous, it attains a minimum value ℓ and maximum value u on $[a, b]$. The average value

$$\overline{f} = \frac{1}{b-a} \int_a^b f(x) \mathrm{d}x$$

can be bounded from below by

$$\frac{1}{b-a} \ell (b-a) = \ell$$

and from above by

$$\frac{1}{b-a} u (b-a) = u$$

The intermediate value theorem then implies that f attains the value $\overline{f}$ between where it attains its minimum and where it attains its maximum. □

2.3 The Mean Value Theorem for Derivatives

The fundamental theorem of calculus connects the mean value theorem for integrals to the mean value theorem for derivatives, which may be more familiar to you.

Theorem 3 (Mean Value Theorem for Derivatives)**.** *If f is a continuous function on $[a, b]$ which is differentiable on (a, b), then there is an $x_0 \in (a, b)$ such that*

$$f'(x_0) = \frac{f(b) - f(a)}{b-a}$$

This statement has some consequences which we have seen before, but which I will reiterate now because we will use them in our next topic.

Proposition. *Let f be differentiable on some interval.*

1. *If $f' > 0$ on the interval, then f is strictly increasing: whenever $x_1 < x_2$, $f(x_1) < f(x_2)$.*
2. *If $f' < 0$ on the interval, then f is strictly decreasing: whenever $x_1 < x_2$, $f(x_1) > f(x_2)$.*
3. *If $f' = 0$ on the interval, then f is constant.*

Proof of (1). Choose $x_3 \in (x_1, x_2)$ such that

$$f'(x_3) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Then $f(x_2) - f(x_1) = (x_2 - x_1) \cdot f'(x_3) > 0$. □

Proofs of the other two statements are similar.

We technically used statement (3) in our proof of the fundamental theorem of calculus, so I should point out that this mean value theorem can be proved in a different without using the fundamental theorem of calculus. See Chapter 4 of the book for how this is done.

Interlude on Transcendental Functions

So far, we have studied some interpretations and applications of integrals using the following tools for evaluating integrals:

1. Antiderivatives of certain simple functions
2. Substitution

This has gotten us pretty far, but we still cannot find explicit integrals for many functions; in particular we still cannot integrate x^{-1}. In the second part of the course, we will see more techniques of integration that allow us to evaluate more integrals. In general, however, to express more general integrals we will need to introduce new functions, beyond the scope of the algebraic functions thus far considered. These transcendental functions will be the next main topic of the course.

Before we get into this, I want to say precisely what it means for a function to be algebraic or transcendental.

Definition. A function f is *algebraic* if there are polynomials $p_0, \dots, p_n$, not all zero, such that the equation

$$p_n(x)f(x)^n + \cdots + p_1(x)f(x) + p_0(x) = 0$$

holds where f is defined. In other words, f satisfies a polynomial equation whose coefficients are polynomial functions of x. A function which is not algebraic is said to be transcendental.[2]

Example 14. A rational function $f(x) = p(x)/q(x)$ satisfies $q(x)f(x) - p(x) = 0$.

Example 15. The nth root function $f(x) = x^{1/n}$ satisfies $f(x)^n - x = 0$.

Remark. It turns out that any function which can be built up using rational functions and nth roots by additional, multiplication, and composition is algebraic. This is not at all clear, it takes some work to construct the associated polynomial equations. Not all algebraic functions can be built up in this way, however. The Bring radical $B(x)$, which satisfies $B(x)^5 + B(x) + x = 0$, is an algebraic function which cannot be expressed in this way.

So far the only transcendental functions we have seen are the trigonometric functions $\sin(x)$ and $\cos(x)$, which can be defined geometrically. Again, it is not at all clear how to see that such functions are transcendental, but there are ways of showing this, outside the scope of this course.

3 Inverses of Functions

One way we will introduce new functions is as inverses of ones we already understand, so our first task will be to clarify when inverse functions exist, and what properties they have.

3.1 Injectivity, Surjectivity, and Inverses

Here I will diverge from the language of the textbook a bit. When thinking about inverses of functions, it is helpful to specify what the inputs and outputs of the function are. We can write something like $f : \mathbb{R} \to \mathbb{R}$ to specify that f is defined for all real numbers and outputs real numbers.

Example 16. The expression

$$f(x) = \frac{x^2 + 1}{(x-1)^2}$$

defines a function $f : (-\infty, 1) \cup (1, \infty) \to [1, \infty)$. We could also view it as a function $f : (-\infty, 1) \cup (1, \infty) \to \mathbb{R}$.

[2] Compare the definition of an algebraic number: one which satisfies a polynomial equation with rational coefficients. A transcendental number is one which is not algebraic.

Definition. Given a function $f : X \to Y$, and inverse of f is a function $g : Y \to X$ such that $g(f(x)) = x$ for all $x \in X$ and $f(g(y)) = y$ for all $y \in Y$.

Example 17. The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x$ has inverse $g : \mathbb{R} \to \mathbb{R}$ defined by $g(x) = x/2$.

An equivalent statement of the definition of inverse is that $g(y)$ is the unique solution in X to the equation $f(x) = y$, for any y. This characterization is useful for two reasons. First, it tells us how to find a formula for an inverse: set up the equation $f(x) = y$ and solve for x in terms of y. Second, it tells us what needs to be true for an inverse to exist.

Definition. A function $f : X \to Y$ is *injective*[3] if whenever $x_1 \neq x_2$ are distinct elements of X, $f(x_1) \neq f(x_2)$. Equivalently, if $f(x_1) = f(x_2)$, then it must be that $x_1 = x_2$. This is to say, equations $f(x) = y$ have at most one solution x.

Definition. A function $f : X \to Y$ is *surjective* if for any $y \in Y$, there is an $x \in X$ with $f(x) = y$. That is, the equation $f(x) = y$ always has a solution.

Proposition. *A function has an inverse if and only if it is injective and surjective. If a function has an inverse, the inverse is unique.*

Example 18. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$. This function is neither injective nor surjective: the equation $f(x) = -1$ has no solution in $\mathbb{R}$, and we have $f(-1) = f(1)$. If we view f as a function $f : \mathbb{R} \to [0, \infty)$, it is now surjective, as we only consider nonnegative output, but it is still not injective. If we restrict f to a function $f : [0, \infty) \to [0, \infty)$, it is now injective and surjective, and so it has an inverse: the square root function $g(x) = \sqrt{x}$, defined as a function $[0, \infty) \to [0, \infty)$.

Example 19. Let's find an inverse for the function $f(x) = 2x^3 + 1$, which is injective and surjective as a function $\mathbb{R} \to \mathbb{R}$. We set up the equation $y = 2x^3 + 1$ and solve for x in terms of y:

$$\rightsquigarrow 2x^3 = y + 1 \rightsquigarrow x^3 = \frac{y+1}{2} \rightsquigarrow x = \left(\frac{y+1}{2}\right)^{1/3}$$

so our inverse is $g(y) = \left(\frac{y+1}{2}\right)^{1/3}$.

Henceforth, we write f^{-1} for the inverse of f, if it exists.

3.2 Criteria for Invertibility

Using the techniques of calculus, we have some useful criteria for a function to be injective or surjective.

Proposition. *Suppose that $f : \mathbb{R} \to \mathbb{R}$ is differentiable, and that $f'(x)$ is either everywhere positive or everywhere negative. Then f is injective.*

Proof. In this case, f is either strictly increasing or strictly decreasing, so whenever $x_1 < x_2$ are two distinct inputs, $f(x_1) < f(x_2)$ (or alternatively $f(x_1) > f(x_2)$), so it cannot be that $f(x_1) = f(x_2)$. □

Example 20. The function

$$f(x) = x + \frac{x^3}{1+x^2}$$

has

$$f'(x) = 1 + \frac{3x^2(1+x^2) - x^3(2x)}{(1+x^2)^2} = 1 + \frac{3x^4 + 3x^2 - 2x^4}{(1+x^2)^2} = 1 + \frac{x^4 + 3x^2}{(1+x^2)^2}$$

This is ≥ 1 everywhere, so f is strictly increasing and therefore injective.

Remark. Actually, we only need that f' is everywhere nonnegative and is not zero on any interval. [Draw picture with x^3]

[3] the book uses "one to one"

Proposition. *Suppose that $f : \mathbb{R} \to \mathbb{R}$ is continuous. If for any $a > 0$, f takes values greater than a and less than $-a$ (this is to say, f is unbounded in both directions), then f is surjective.*

Proof. For any b, Choose a so that $-a < b < a$. Then apply the intermediate value theorem. [Draw picture.] □

3.3 The graph of the inverse function

The graph of a function f consists of all points $(x, f(x))$. If f has an inverse, then its graph consists of points $(x, f^{-1}(x))$. Writing $y = f^{-1}(x)$, we have $x = f(y)$, so the graph consists of points $(f(y), y)$. This is to say, it is obtained from the graph of f by switching the x and y coordinates. Geometrically, this corresponds to reflecting the graph over the line $y = x$ [Draw picture.]

Example 21. Graph $x^3 + 1$ and $(x - 1)^{1/3}$.

3.4 The derivative of the inverse function

If a function is continuous and differentiable, and has an inverse, then its inverse inherits these properties, at least partially.

Proposition. *Suppose that f is a continuous injective function from an interval I to an interval J. Then $f^{-1} : J \to I$ is continuous.*

If f^{-1} is also differentiable, we can find what its derivative must be. For notational simplicity, write $g = f^{-1}$, and consider the equation:

$$f(f^{-1}(x)) = x$$

Taking the derivative of both sides, using the chain rule, we obtain:

$$f'(g(x))g'(x) = 1$$

and therefore

$$g'(x) = \frac{1}{f'(g(x))}$$

at least if both sides make sense. It turns out that the left side makes sense wherever the right side does.

Proposition. *If f has inverse $g = f^{-1}$, and $f'(g(a))$ exists and is nonzero, then $g'(a)$ exists and is equal to $1/f(g(a))$.*

Remark. Because the derivative of a function is a "local" construction, depending only on the function near the point, g only needs to be an inverse of f on some interval containing a.

Example 22. Let's consider the case of an odd power $f(x) = x^n$. This is invertible as a function $\mathbb{R} \to \mathbb{R}$, with inverse $g(y) = y^{1/n}$. We already know what the derivatives of both of these functions are, but let's compute $g'(y)$ using the above formula to make sure that it works. We have $f'(x) = nx^n$, and so

$$\frac{1}{f'(g(y))} = \frac{1}{n(y^{1/n})^{n-1}} = \frac{1}{n}y^{-\frac{n-1}{n}} = \frac{1}{n}y^{\frac{1}{n}-1}$$

This agrees with the power rule for fractional powers. Notice that this expression is undefined for $y = 0$. Indeed, $y^{1/n}$ is not differentiable here, the graph has a vertical tangent line. [Draw picture.] This corresponds to a horizontal tangent line to the graph of x^n.

Later on, we will apply these considerations to inverse trigonometric functions. Before that, though, we will introduce the logarithm and exponential functions.